Strategic Math Thinking – Student Version

Instructor: Adam Earnst

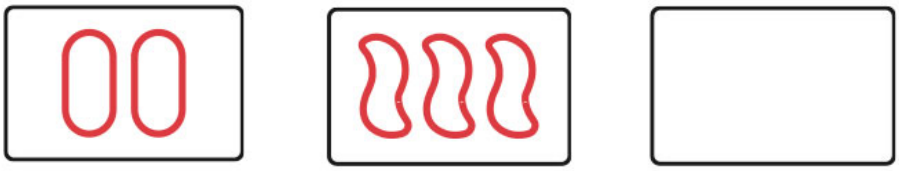
July 8 - 25, 2025

Course Overview

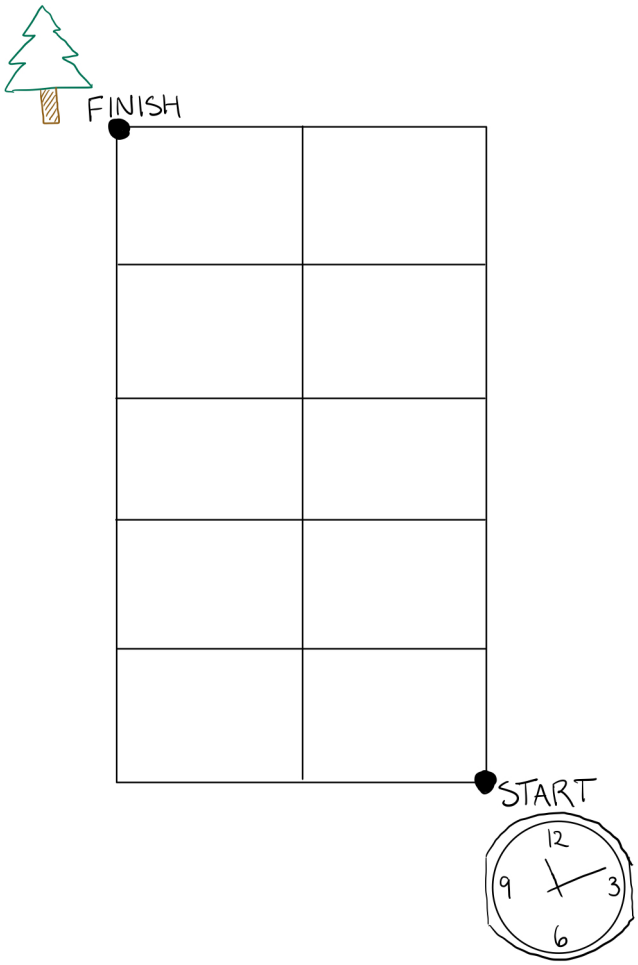
Title: Strategic Math Thinking

Description: This course is all about finding creative ways to solve problems. Sure, we all know how to count to 10, to 100, and so on. But what if you wanted to count how many different outfits you could assemble from your entire wardrobe? That would be a much harder problem! In this course, we'll develop tools that will help us approach problems like this (and solve them quickly, too). Throughout this class, we'll use the knowledge we gain to investigate the math behind the popular game, SET!

Pitch: We're going to play a game.



Sample Problem: Albert just arrived at Grand Central Station in New York City! After seeing the clock at the station, he wants to go to the giant Christmas tree at Rockefeller Center. He wants to get there quickly, so he wants to take a route that only goes up and to the left, one block at a time. (Example: one path that works is L L U U U U U). How many different paths can Albert take?


FINISH
START
12
3
6
9

Course Norms

- I will be excellent to others.
- I will be here to grow.
- I will keep myself and others safe.
- I will listen to staff instructions.

Now let's add our own course norms:

_____

_____

_____

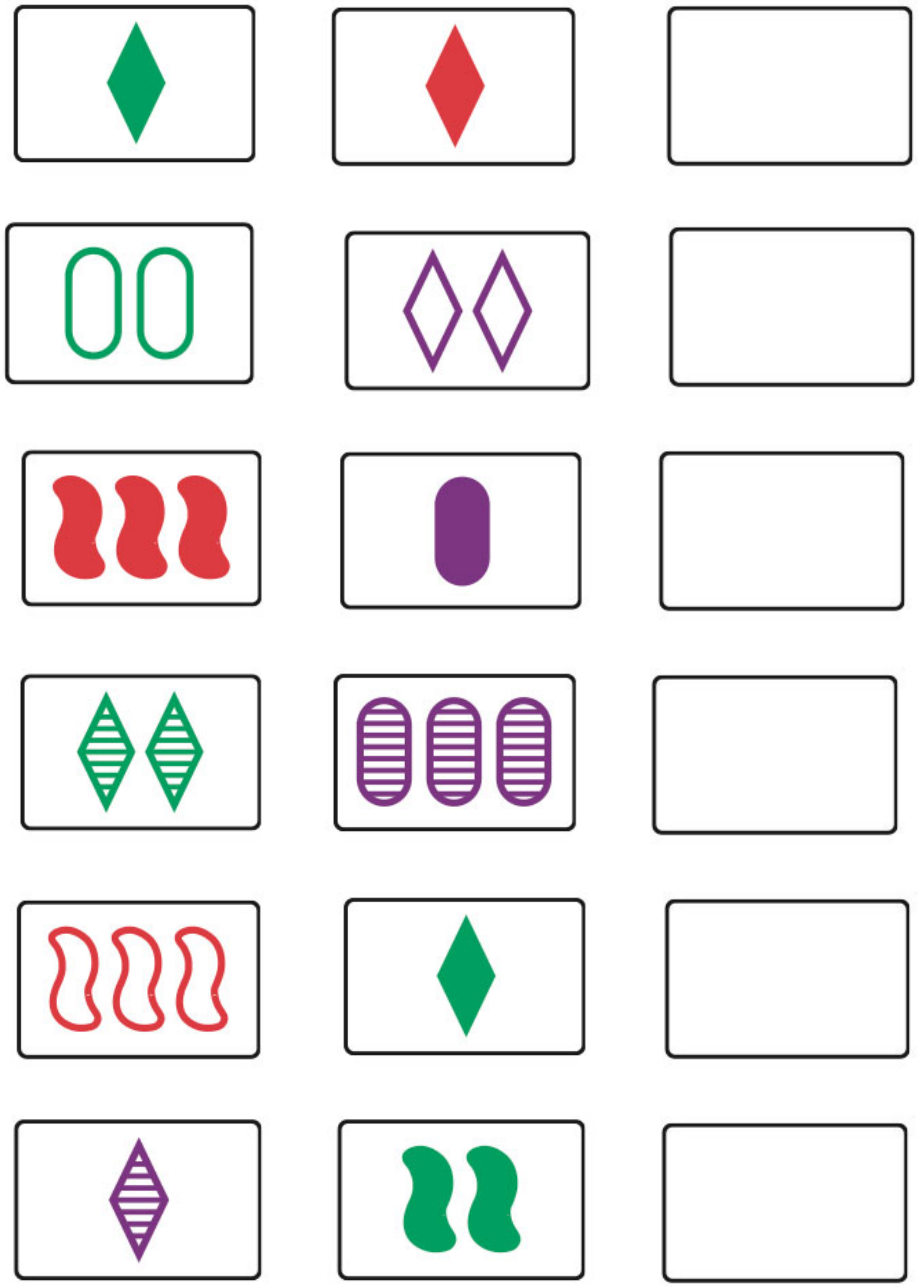
_____

_____

Day 1

What's the missing card?

Below are rows of three cards, but one is empty in each row. What card is missing to complete the SET?



If you're done, what's are the similarities and differences between the sets above? Can we group them into categories?

Let's play SET!

You can write any observations in the space below.

How many cards are in the game of SET?

And what about the game of T∃S? (don't worry, I'll explain what this is!)

Week 1 Reflections

We will use this sheet at the end of every class to write down some key takeaways from that day!

Day	**One thing that was fun**	**One thing I learned**	**One question I have**
Day 1			
Day 2			
Day 3			
Day 4			
Strategies I used this week:			

Day 2 - 3

Ice Cream Assembly

You are going to an ice cream shop which has three flavors: chocolate, vanilla, and pistachio. You can get two scoops on your cone. How many different cones can you make?

Solution 1: order matters	Solution 2: order doesn't matter

Now, let's assume that order matters. What if you can get three scoops? What if you can get four?

Counting With Order

License Plates: New York license plates are typically 3 letters followed by 4 numbers. How many different license plates are there?

Answer: _____

Ryan's fancy license plate: Ryan wants to get a New York license plate where no two consecutive characters are the same. How many choices does he have now?

Answer: _____

Drink of the day: There are 48 drink options in the dining hall. Michael wants to create a drink using at least one of the 48 options. How many different drinks can he create?

Answer: _____

More with TƎS: How many ways can we create a 3x3 grid of TƎS cards such that every row, column, and diagonal goes together?

Answer: _____

Forwards and backwards: A palindrome is a number that reads the same forwards and backwards (like 12321).

(a) How many 4-digit palindromes are there?

(b) How many 5-digit palindromes are there?

(c) How many 6-digit palindromes are there?

(d) Do you spot a pattern? How many n-digit palindromes are there?

Day 3

Building Towers

How many ways can you build a tower with three different colored blocks?

What about for four different colored blocks?

What about five? Do you notice a pattern?

Can you come up with a formula to solve this problem for n different colored blocks?

Mixed Drinks

The fountain in the dining hall has 48 different drink options. How many different drinks can you create by mixing together at least one of the drink options? [Hint: this is a hard problem - use your toolkit of problem-solving strategies!]

Day 4

Writing a Combinatorial Proof

We have a soda fountain which contains n different drink types. We want to figure out how many different drinks we can mix together.
For example, with two drink options, which we can call drink A and drink B, we have three options:

A
B
AB

With three drink options, which we can call drink A, drink B, and drink C, we have seven options:

A
B
C
AB
AC
BC
ABC

Can you list all the options if we have 4 drinks, in a similarly organized way? What about 5? Write on your chart paper!

Drink Options	**Number of Mixed Drinks**	**Number of Mixed Drinks (including empty)**
1	1	2
2	3	4
3	7	8
4		
5		

Can we make a **conjecture** about the number of mixed drinks (including the empty drink) if we have n drink options?

Let Adam or your TA know when you have a conjecture.

Let's use the symbol $\varnothing$ to represent the empty drink. So, we can list the mixed drink options including the empty drink. I've organized them in a different way this time.

1 Drink (A):	$\varnothing$				A			
2 Drinks (A, B):	$\varnothing$	B			A	AB		
3 Drinks (A, B, C):	$\varnothing$	C	B	BC	A	AC	AB	ABC

Does this look like a type of diagram we've used before? What would the row for 4 drinks look like? What about 5? Draw on your chart paper!

Now, can you **prove** your conjecture? Convince Adam or your TA!

Math Terms

Definition: A precise description of a mathematical term or idea.

- An even number is an integer that can be expressed as $2 \times n$ for another integer n.
- A pentomino is a shape formed by connecting five squares of the same size edge-to-edge.
- A proposition is a statement that is either true or false.

Conjecture: A mathematical statement that someone believes is true - an educated guess.

- I believe that every even number greater than 2 can be written as the sum of two prime numbers.

Theorem: A true mathematical statement that has been proven.

- There are infinitely many prime numbers.

Proof: An argument that carefully shows that a mathematical statement is always true.

Counterexample: An example that shows you that a mathematical statement is incorrect.

- Suppose Adam believes that all prime numbers are odd. This is a false statement! The number 2 is a counterexample, because it is prime and even.

A soda fountain has n different drink options. A mixed drink is any combination of zero or more drinks mixed together. Two mixed drinks are considered the same if they have the same ingredients - amounts do not matter.

Trying this problem for small values of n, I see a pattern. I think the number of mixed drinks is 2^n.

I drew a tree diagram and explained why the number of mixed drinks doubles every time n increases.

If a soda fountain has n different drink options, there are 2^n different mixed drinks, including the empty drink.

Day 5

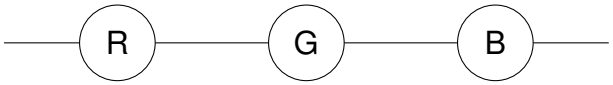
Counting Bracelets

Question 1: I have a bracelet string and beads of different colors.

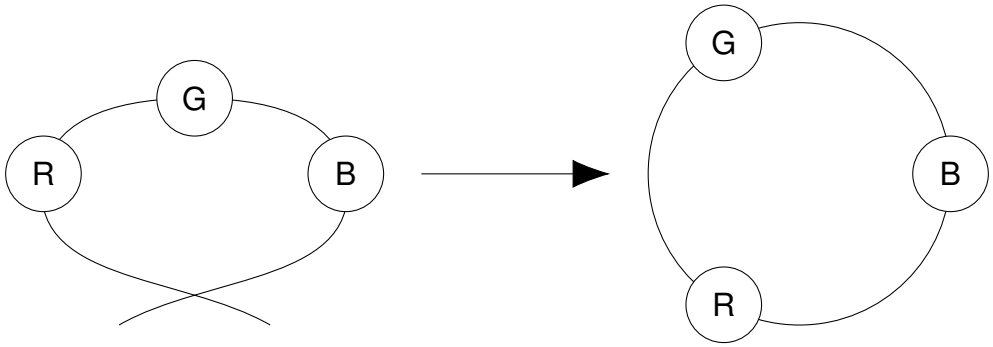
(a) How many different ways can I order a red and blue bead on a string?

(b) How many different ways can I order a red, green, and blue bead on a string?

I've written down one possible arrangement - make sure to draw them all even if you can calculate it otherwise! We'll need this for the next question.



Question 2: I want to tie these strings with beads to create a bracelet. I do this by bringing the ends of the string down, tying them together, and spacing the beads out evenly.



Do this procedure for the strings you drew in question 1. You will get some bracelets that can be rotated to get others in your list. Group all of those bracelets together. How many groups do you end up with? How many bracelets are in each group?

Question 3: If you finished Questions 1 and 2, try doing the same question for four beads: red, green, blue, and yellow.

Can you complete the table below?

Bead colors	**Number of line arrangements**	**Number of line arrangements in each group**	**Number of bracelet arrangements**
1			
2			
3			
4			

Question 4: Do you see a pattern? Can you come up with a conjecture about the number of bracelet arrangements for n different bead colors.

Question 5: Try to prove your conjecture. Ask your friends and teachers for help!

Super-Challenge Question: How many bracelet arrangements are there if some of the beads are the same color? As an example, try 2R, 1G, 1B. If you figure this out, what about 2R, 1G, 1B, 1Y?

Week 2 Reflections

We will use this sheet at the end of every class to write down some key takeaways from that day!

Day	**One thing that was fun**	**One thing I learned**	**One question I have**
Day 5			
Day 6			
Day 7			
Strategies I used this week:			

Day 6

Permutations and Combinations

How many ways can you choose a president and vice-president of your group? Does order matter for this question?

In your groups, each pair of people needs to high five. How many introductions does it take for everyone to say hi to everyone else? Does order matter for this question?

How do these answers compare to each other?

How many ways can you choose a president, vice-president, and secretary of your group? Does order matter here?

How many different ways can you pick 3 people from your group to play SET? Does order matter here?

How do these answers compare to each other?

Combinations and SET

How many ways can we choose 2 objects from a group of n?

How many ways can we choose 3 objects from a group of n?

Question 1: How many different triplets of SET cards are there? [Hint: does order matter or not?]

Question 2: This is a hard problem, so we'll do it in steps. The goal is to figure out how many different SETs there are in the game of SET. We'll start by asking this question for the game of TƎS.

(a) Given a pair of TƎS cards, how many cards are there that complete the set?

(b) How many different pairs of TƎS cards are there?

(c) Can you use the answers from (a) and (b) to guess how many sets there are in TƎS?

(d) Try to list out all of the possible sets in TƎS to check your guess. I promise there aren't many! How is this answer similar or different from your guess?

Day 7

See handwritten notes. Counted number of SETs in SET.

Day 8

Game 1: Evens or Odds

Directions: The game starts with a pile of 15 chips in the center of the table.

Each round, both players roll one die. Add the results together.

If the result is odd, player 1 takes a chip from the center. If the result is even, player 2 takes a chip from the center.

The player with the most chips after there are no chips left in the center of the table wins.

Game 2: Green Diamonds

Directions: Shuffle the deck of SET cards.

Starting with player 1, the two players alternate drawing cards from the top of the deck.

The first player to draw a card with green diamonds wins.

Game 3: Evens or Odds 2.0

Directions: The game starts with a pile of 15 chips in the center of the table.

Each round, both players roll one die. Multiply the results together.

If the result is odd, player 1 takes **two** chips from the center. If the result is even, player 2 takes **one** chip from the center.

The player with the most chips after there are no chips left in the center of the table wins.

Game 4: Evens and Threevens

Directions: The two players alternate rolling a die, starting with player 1.

Player 1 wins if they roll a **threeven** number (a number divisible by 3). Player 2 wins if they roll an even number.

Neither player can win on the other player's roll.

Game 5: Prime Dice

Directions: In the middle of the table, there are a collection of different dice and 15 chips.

The first phase of this game is die selection. Player 2 starts, picking a die from the table. Then player 1 picks one of the remaining dice.

Then, both players roll their die at the same time. If a player's die lands on a prime number, take a chip from the center. If both players dice land on a prime but there is only one chip in the center, re-roll.

The winner is the player with the most chips after there are none left in the center.

Game 6: Contract Dice

Directions: This game begins with a sign-up phase. Start by writing the numbers 2 through 12 on the whiteboard.

Players take turns signing up for numbers, starting with player 1. To sign up for a number, write your name next to that number.

When there are only two numbers left, player 2 gets to sign up for both of them, to compensate for not getting the first pick.

Then, the rolling phase begins: each player rolls a die and the results are added together. Whoever signed up for the resulting number takes a chip from the center.

The winner is the person who has the most chips when there are none left in the center.

Scorecard

Player 1 ☐
Player 2 ☐

Game	Games won	Games lost	Space for any other notes
Game 1: Evens or Odds			
Game 2: Green Diamonds			
Game 3: Evens or Odds 2.0			
Game 4: Evens and Threevens			
Game 5: Prime Dice			
Game 6: Contract Dice			

Day 9

Create your own fair/unfair game

Day 10

SKUNK

Question 1: What is the probability of getting snake eyes on a roll of two dice? (Snake eyes occurs when both dice are showing a 1).

Question 2: What is the probability of getting a SKUNK on a roll of two dice? (A SKUNK occurs when at exactly one of the two dice is showing a 1 on its face).

Question 3: What is the probability of getting a SKUNK or snake eyes on a roll of two dice?

Question 4: What is the probability of NOT getting a SKUNK or snake eyes on a roll of two dice?

Question 5: What is the average score across all rolls where you don't get a SKUNK or snake eyes?

Expected Value

$$\text{Expected Value} = \sum \text{probability} \times \text{amount won or lost.}$$

As an example, suppose you in the first round of SKUNK, and you have a score of 50 points so far.
If a SKUNK or snake eyes are rolled, you lose 50 points. Otherwise, you gain the points on the two rolled dice. So the expected value of this roll is

$$\begin{aligned}&(\text{probability of SKUNK or snake eyes}) \times (-50)\\ +&(\text{probability of no SKUNK or snake eyes}) \times (\text{average score where no SKUNK/snake eyes}).\end{aligned}$$

Question 6: What is the expected value of this roll? Should you sit down before the dice or rolled or stay standing up?

Question 7: Try to come up with a good strategy for SKUNK.

SKUNK Scorecard 1

S	K	U	N	K

SKUNK Scorecard 2

S	K	U	N	K

SKUNK Scorecard 3

S	K	U	N	K

SKUNK Scorecard 4

S	K	U	N	K

SKUNK Scorecard 5

S	K	U	N	K

SKUNK Scorecard 6

S	K	U	N	K

SKUNK Scorecard 7

S	K	U	N	K

SKUNK Scorecard 8

S	K	U	N	K

SKUNK Scorecard 9

S	K	U	N	K

SKUNK Scorecard 10

S	K	U	N	K

SKUNK Scorecard 11

S	K	U	N	K

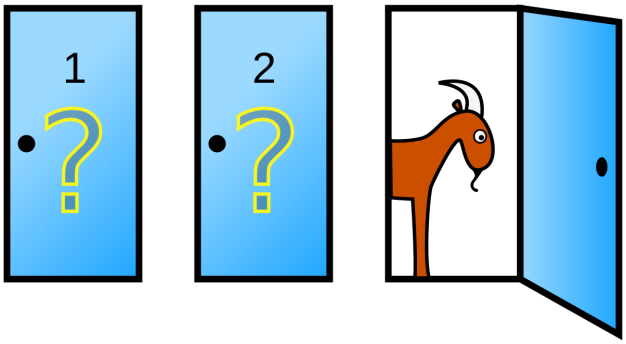
SKUNK Scorecard 12

S	K	U	N	K

Day 11

Cars 'n' Goats

You are in a gameshow, and the host shows you three closed doors. She says that behind one of the doors is a brand new car. Behind the other two are goats (a very confusing prize). She asks you to pick a door. After picking a door, she reveals a goat behind one of the doors you didn't pick. She then gives you the option to switch your door choice. You win whatever is behind your door.



Stay or Switch?

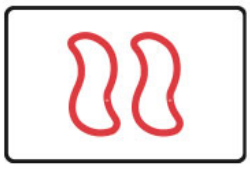
Game Number	Stay or Switch?	Win or Lose?
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Question: Which strategy seemed to work better (or were they equally good)? Can you try to figure out why?

Secret SET

1100

1210



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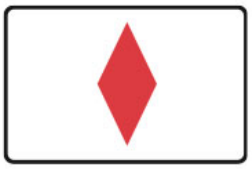

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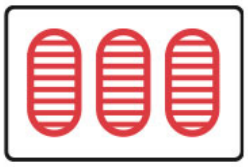

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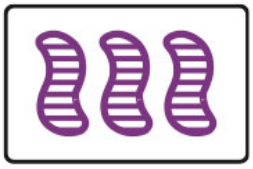
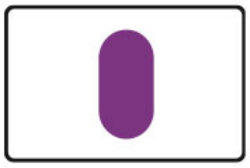


2221

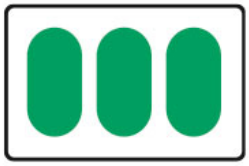
0122



0202

2111

1112



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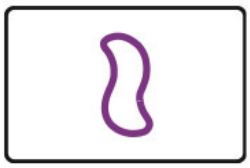

0220

Question: Find as many SETs as you can! How many do you count?

Question: Can you decode the secret numbers listed below each card? Hint: Look for any patterns you can find!!

Question: What do you notice about the secret numbers for cards that form a SET? [if you're stuck, ask Adam for a hint!]